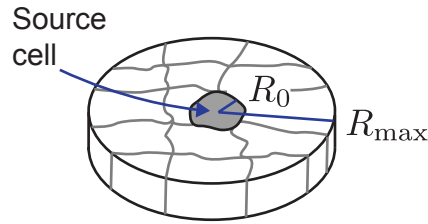


Problem Set 3: Morphogens in an epithelial sheet

Physics 230A/146A — Biological Physics

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Many tissues are comprised of two-dimensional sheets of cells, including epithelial sheets that line the lungs, gut, and blood vessels. The cells in these tissues communicate by emitting and receiving small diffusive molecules called ligands.



For mathematical convenience, assume the epithelial sheet is an annulus (circular disk with a circular hole removed from the center) with inner radius R_0 and outer radius $R_{\max}$. Assume that the ligand diffuses only along the two-dimensional surface, so that

$$\frac{\partial c}{\partial t} = D \nabla^2 c \quad R_0 < r < R_{\max}$$

where in two dimensions

$$\nabla^2 c(r, \theta) = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 c}{\partial \theta^2}.$$

Suppose the source of diffusible ligand is emitted from the center ring at $r = R_0$, so that

$$\begin{aligned}J(R_0, t) &= J_0, \\c(R_0, t) &= c_0\end{aligned}$$

where $J(r, t)$ is the flux in the radial direction. Assume $J_0 > 0$ so that it is, indeed, a source of ligand.

Find the steady-state ligand concentration under the following assumptions:

- i. If the ligand does not bind (or binds and then detaches rapidly) from the other cells in the sheet
- ii. If the ligand binds irreversibly to other cells at a constant internalization rate, $d(r, t) = \gamma$

In the first case, (i), is the flux at $r = R_0$ equal to the flux at $r = R_{\max}$?